A Formal Proof of the Conservation of the Laplace–Runge–Lenz Vector and Kepler's First Law in Lean 4

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Abstract

We present a machine-checked formalization, in the Lean 4 proof assistant on top of the Physlib library, of the classical Kepler problem: a particle in the inverse-square potential $V(r) = -k/r$. We define the Laplace–Runge–Lenz (LRL) vector on phase space and prove that its Poisson bracket with the Hamiltonian vanishes away from the centre of force. We prove the algebraic identities $\langle \mathbf{A}, \mathbf{L} \rangle = 0$, $\langle \mathbf{A}, \mathbf{q} \rangle = \|\mathbf{L}\|^2 - mk\|\mathbf{q}\|$ and $\|\mathbf{A}\|^2 = (mk)^2 + 2mH\|\mathbf{L}\|^2$, and derive that bound orbits ($H < 0$) have eccentricity less than one. Finally, we lift these phase-space statements to trajectories: along every differentiable solution of Hamilton's equations avoiding the origin, the LRL vector and $\|\mathbf{L}\|^2$ are constant, so the position satisfies the polar equation of a conic with a focus at the centre of force (Kepler's first law). The development is about 300 lines, uses no axioms beyond Lean's standard three, and passes all of Physlib's contribution linters.

1 Introduction

Formal verification of physics has gained momentum with the growth of Lean's mathematical library Mathlib [3] and the physics library Physlib (formerly HepLean/PhysLean) [4]. Physlib now contains, among much else, a quantum-mechanical treatment of the hydrogen atom including commutation relations of the quantum LRL operator, and, as of October 2026, basic infrastructure for classical central-force problems: the Hamiltonian $H(\mathbf{p}, \mathbf{q}) = \langle \mathbf{p}, \mathbf{p} \rangle / 2m + V(\|\mathbf{q}\|)$, its gradients, and the conservation of angular momentum in Poisson-bracket form.

What was missing is the defining classical result of the Kepler problem: the existence of the additional conserved vector $\mathbf{A}$ that explains why Kepler orbits close, and the derivation of the conic orbit equation from it. Before starting we searched Physlib's sources, its open pull requests and issues, and the web; we are not aware of an existing formalization of the classical LRL vector or of the conic orbit theorem in Lean. This note fills that gap.

Contributions.

- A definition of the Kepler problem as an instance of Physlib's `CentralForce` structure, and of the LRL vector as a function on phase space (§3.1).

*AI system (model Claude Opus 5.5), developed by Anthropic. Claude wrote the Lean proofs and drafted this manuscript in collaboration with the first author.

- Algebraic identities: the triple-product expansion of $\mathbf{A}$, orthogonality to $\mathbf{L}$, the orbit equation, the norm identity, and the eccentricity–energy criterion (§3.2).
- Conservation: explicit phase-space gradients of each component of $\mathbf{A}$ and the vanishing of $\{A_i, H\}$ (§3.3).
- Trajectory results: a general lemma that a Poisson-commuting observable is constant along solutions of Hamilton's equations, and Kepler's first law (§3.4).

The source is available in the `kepler-lrl` folder of `https://github.com/Crotalus/ai-formalized`; it builds against Physlib commit `3bfc7af` with Lean `v4.35.0`.

2 Mathematical background

Consider a particle of mass $m > 0$ at position $\mathbf{q} \in \mathbb{R}^3$ with momentum $\mathbf{p}$ in the potential $V(r) = -k/r$, with Hamiltonian

$$H(\mathbf{p}, \mathbf{q}) = \frac{\langle \mathbf{p}, \mathbf{p} \rangle}{2m} - \frac{k}{\|\mathbf{q}\|}.$$

The angular momentum is $\mathbf{L} = \mathbf{q} \times \mathbf{p}$ and the Laplace–Runge–Lenz vector is

$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - \frac{mk}{\|\mathbf{q}\|}\,\mathbf{q} = \langle \mathbf{p}, \mathbf{p} \rangle\,\mathbf{q} - \langle \mathbf{q}, \mathbf{p} \rangle\,\mathbf{p} - \frac{mk}{\|\mathbf{q}\|}\,\mathbf{q},$$

where the second form uses $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \langle \mathbf{a}, \mathbf{c} \rangle \mathbf{b} - \langle \mathbf{a}, \mathbf{b} \rangle \mathbf{c}$ [1, §3.9]. Taking the inner product with $\mathbf{q}$ and using Lagrange's identity $\|\mathbf{L}\|^2 = \langle \mathbf{q}, \mathbf{q} \rangle \langle \mathbf{p}, \mathbf{p} \rangle - \langle \mathbf{q}, \mathbf{p} \rangle^2$ gives

$$\langle \mathbf{A}, \mathbf{q} \rangle = \|\mathbf{L}\|^2 - mk\|\mathbf{q}\|. \tag{1}$$

If $\mathbf{A}$ and $\mathbf{L}$ are constants of motion and θ denotes the angle between $\mathbf{A}$ and $\mathbf{q}$, (1) reads $\|\mathbf{q}\|(mk + \|\mathbf{A}\| \cos\theta) = \|\mathbf{L}\|^2$, that is $\|\mathbf{q}\| = \frac{\|\mathbf{L}\|^2/mk}{1 + e\cos\theta}$ with eccentricity $e = \|\mathbf{A}\|/mk$: a conic with focus at the origin. A direct expansion also gives

$$\|\mathbf{A}\|^2 = (mk)^2 + 2mH\|\mathbf{L}\|^2, \tag{2}$$

so for $k > 0$ and $\mathbf{L} \neq 0$, $e < 1$ if and only if $H < 0$. Elementary derivations of these facts can be found in [1, 2].

3 Formalization

3.1 Setting and conventions

Physlib models phase space for a particle in three dimensions as pairs $(\mathbf{p}, \mathbf{q})$ of elements of `EuclideanSpace` $\mathbb{R}$ `(Fin 3)`, and observables as curried functions `p` $\to$ `q` $\to \mathbb{R}$. The Poisson bracket is

$$\{f, g\} = \langle \nabla_{\mathbf{q}} f, \nabla_{\mathbf{p}} g \rangle - \langle \nabla_{\mathbf{p}} f, \nabla_{\mathbf{q}} g \rangle,$$

and a central-force system is a structure carrying a mass $m > 0$ and a potential `V` : $\mathbb{R} \to \mathbb{R}$ smooth on $(0, \infty)$. We follow these conventions exactly so that the development can be contributed upstream. The Kepler problem and the LRL vector are:

```
noncomputable def kepler (m k : ℝ) (hm : 0 < m) : CentralForce where
  m := m
  V := fun r => -k / r
  m_pos := hm
  contDiffOn_V := contDiffOn_const.div contDiffOn_id
    fun _ hr => (Set.mem_Ioi.mp hr).ne'

noncomputable def lrlVector (m k : ℝ) (p q : EuclideanSpace ℝ (Fin 3)) :
    EuclideanSpace ℝ (Fin 3) :=
  WithLp.toLp 2 (p.ofLp ×₃ (angularMomentum p q).ofLp) - (m * k / ‖q‖) • q
```

The definition of `lrlVector` mirrors the textbook formula $\mathbf{p}\times\mathbf{L}-(mk/\|\mathbf{q}\|)\mathbf{q}$, reusing Physlib's `angularMomentum` and Mathlib's cross product on `Fin 3` $\to \mathbb{R}$. The `toLp`/`ofLp` coercions are needed because the cross product lives on the plain function type, whereas norms and inner products live on `EuclideanSpace`.

Remark 1. Lean's convention $x/0 = 0$ makes `lrlVector` total. A consequence we exploited is that the orbit equation (1) holds *without* the hypothesis $\mathbf{q} \neq 0$: at $\mathbf{q} = 0$ both sides vanish. In contrast, (2) genuinely requires $\mathbf{q} \neq 0$.

3.2 Algebraic identities

```
lemma lrlVector_eq (m k : ℝ) (p q : EuclideanSpace ℝ (Fin 3)) :
    lrlVector m k p q = ⟪p, p⟫_ℝ • q - ⟪q, p⟫_ℝ • p - (m * k / ‖q‖) • q

lemma inner_lrlVector_angularMomentum (m k : ℝ) (p q : EuclideanSpace ℝ (Fin 3)) :
    ⟪lrlVector m k p q, angularMomentum p q⟫_ℝ = 0

lemma inner_lrlVector_position (m k : ℝ) (p q : EuclideanSpace ℝ (Fin 3)) :
    ⟪lrlVector m k p q, q⟫_ℝ = angularMomentumSq p q - m * k * ‖q‖

lemma norm_lrlVector_sq (m k : ℝ) (hm : 0 < m) (p : EuclideanSpace ℝ (Fin 3))
    {q : EuclideanSpace ℝ (Fin 3)} (hq : q ≠ 0) :
    ‖lrlVector m k p q‖ ^ 2 =
      (m * k) ^ 2 + 2 * m * (kepler m k hm).hamiltonian p q * angularMomentumSq p q

lemma norm_lrlVector_lt_iff (m k : ℝ) (hm : 0 < m) (hk : 0 < k)
    (p : EuclideanSpace ℝ (Fin 3)) {q : EuclideanSpace ℝ (Fin 3)} (hq : q ≠ 0)
    (hL : angularMomentumSq p q ≠ 0) :
    ‖lrlVector m k p q‖ < m * k ↔ (kepler m k hm).hamiltonian p q < 0
```

The triple-product expansion and the orthogonality to $\mathbf{L}$ are proved componentwise: after `ext i; fin_cases i` the goals become polynomial identities in the six coordinates (with $mk/\|\mathbf{q}\|$ as an opaque atom), closed by `simp` and `ring`. All later results are stated in terms of inner products and norms and use `lrlVector_eq` to avoid coordinates. For (2) we expand $\langle \mathbf{A}, \mathbf{A}\rangle$ bilinearly, rewrite with Physlib's Lagrange identity `angularMomentumSq_eq`, clear the denominator $\|\mathbf{q}\|$ with `field_simp`, and close with `ring`. One subtlety: rewriting $\langle x, x\rangle = \|x\|^2$ must happen *after* bilinear expansion, otherwise it fires on $\langle c\,\mathbf{q}, c\,\mathbf{q}\rangle$ and blocks the expansion. The eccentricity criterion follows from (2) by comparing squares of non-negative reals and a single `nlinarith` call with the hint $m\|\mathbf{L}\|^2 > 0$.

3.3 Conservation of the LRL vector

For each component A_i we compute the partial gradients explicitly:

$$\nabla_{\mathbf{p}} A_i = 2q_i\,\mathbf{p} - p_i\,\mathbf{q} - \langle \mathbf{q}, \mathbf{p}\rangle\,\mathbf{e}_i,$$
$$\nabla_{\mathbf{q}} A_i = \langle \mathbf{p}, \mathbf{p}\rangle\,\mathbf{e}_i - p_i\,\mathbf{p} - \frac{mk}{\|\mathbf{q}\|}\,\mathbf{e}_i + \frac{mk\,q_i}{\|\mathbf{q}\|^3}\,\mathbf{q} \qquad (\mathbf{q} \neq 0).$$

In Lean these are `HasGradientAt` statements, assembled from Physlib's gradient calculus (`hasGradientAt_inner_self`, `hasGradientAt_coord`, `hasGradientAt_norm` and a chain rule `HasDerivAt.comp_hasGradientAt`), followed by `module` or `field_simp` to normalize the resulting linear combination. Using $V'(r) = k/r^2$ (lemma `kepler_deriv_V`) and Physlib's gradients of the Hamiltonian, $\nabla_{\mathbf{p}} H = \mathbf{p}/m$ and $\nabla_{\mathbf{q}} H = (k/\|\mathbf{q}\|^3)\,\mathbf{q}$, the Poisson bracket evaluates to

$$\{A_i, H\} = -k\Big(\frac{p_i}{\|\mathbf{q}\|} - \frac{q_i\langle \mathbf{q}, \mathbf{p}\rangle}{\|\mathbf{q}\|^3}\Big) - \frac{k}{\|\mathbf{q}\|^3}\big(q_i\langle \mathbf{q}, \mathbf{p}\rangle - p_i\|\mathbf{q}\|^2\big) = 0.$$

```
lemma poissonBracket_lrlVector_hamiltonian (m k : ℝ) (hm : 0 < m) (i : Fin 3)
    (p : EuclideanSpace ℝ (Fin 3)) {q : EuclideanSpace ℝ (Fin 3)} (hq : q ≠ 0) :
    poissonBracket (fun p' q' => lrlVector m k p' q' i)
      (kepler m k hm).hamiltonian p q = 0
```

The proof is eleven lines: rewrite the bracket via `poissonBracket_eq_of_hasGradientAt` with the four gradients, expand inner products, and close with `field_simp; ring`.

3.4 From phase space to trajectories

A vanishing Poisson bracket is the infinitesimal statement of conservation. To obtain Kepler's first law we need the integrated statement along an actual motion. Physlib provides Hamilton's equations as `hamiltonEqOp H p q = 0` for curves `p q : Time → EuclideanSpace ℝ (Fin 3)`, and the theorem `deriv_comp_eq_poissonBracket` stating $\frac{d}{dt} f(\mathbf{p}(t), \mathbf{q}(t)) = \{f, H\}(\mathbf{p}(t), \mathbf{q}(t))$ along solutions. We proved a general lemma:

```
lemma eq_of_poissonBracket_eq_zero {H f : E → E → ℝ} {p q : Time → E}
    (heq : hamiltonEqOp (fun _ => H) p q = 0)
    (hp : Differentiable ℝ p) (hq : Differentiable ℝ q)
    (hf : ∀ t, DifferentiableAt ℝ (Function.uncurry f) (p t, q t))
    (hpb : ∀ t, poissonBracket f H (p t) (q t) = 0) (t s : Time) :
    f (p t) (q t) = f (p s) (q s)
```

(here `E` abbreviates `EuclideanSpace ℝ (Fin 3)`). Physlib's `Time` is a structure wrapping a real number rather than $\mathbb{R}$ itself; the proof transports the derivative to $\mathbb{R}$ with the bridge lemma `hasDerivAt_mk_of_differentiableAt` and applies Mathlib's `is_const_of_deriv_eq_zero`. Together with differentiability of A_i away from the origin and the existing conservation of $\|\mathbf{L}\|^2$, this yields:

Theorem 2 (Kepler's first law)**.** *Let $(\mathbf{p}(t), \mathbf{q}(t))$ be a differentiable solution of Hamilton's equations for the Kepler Hamiltonian with $\mathbf{q}(t) \neq 0$ for all t. Then for all times t_0, t,*

$$\langle \mathbf{A}(t_0), \mathbf{q}(t)\rangle = \|\mathbf{L}(t_0)\|^2 - mk\,\|\mathbf{q}(t)\|.$$

```
lemma inner_lrlVector_position_of_hamiltonEqOp (m k : ℝ) (hm : 0 < m)
    {p q : Time → EuclideanSpace ℝ (Fin 3)}
    (heq : hamiltonEqOp (fun _ => (kepler m k hm).hamiltonian) p q = 0)
    (hp : Differentiable ℝ p) (hq : Differentiable ℝ q) (hq0 : ∀ t, q t ≠ 0)
    (t₀ t : Time) :
    ⟪lrlVector m k (p t₀) (q t₀), q t⟫_ℝ =
      angularMomentumSq (p t₀) (q t₀) - m * k * ‖q t‖
```

Combined with `norm_lrlVector_lt_iff`, this shows that a bound Kepler motion with non-zero angular momentum traces an ellipse. Note that we assume, rather than construct, a solution of Hamilton's equations; existence and uniqueness of solutions are not needed for these statements and are not addressed.

4 Discussion

Size and checking. The development consists of 2 definitions and 15 lemmas in about 300 lines, including documentation. No proof exceeds a dozen lines. `#print axioms` on the main results reports only `propext`, `Classical.choice` and `Quot.sound`. The file passes Physlib's module-documentation, alpha-library and style linters.

What was easy. Once the right normal form (`lrlVector_eq`) was available, every identity of the Kepler problem reduced to `field_simp` and `ring`. Physlib's existing gradient API and its Poisson-bracket lemma `poissonBracket_eq_of_hasGradientAt` made the conservation proof essentially a transcription of the pen-and-paper computation.

What required care. The friction points were elaboration issues rather than mathematics: the higher-order unification of `DifferentiableAt.comp` and of the chain rule for gradients needed explicit named arguments; `fun_prop` could not discharge differentiability of $\mathbf{q} \mapsto q_i$ on `EuclideanSpace` without hand-supplied facts; and moving between `Time` and $\mathbb{R}$ required Physlib's bridge lemmas.

Future work. Natural continuations are Kepler's second law (constant areal velocity, essentially the conservation of $\mathbf{L}$ along trajectories), Kepler's third law relating the period to the semi-major axis, the hidden $SO(4)$ symmetry via the Poisson algebra of $\mathbf{L}$ and $\mathbf{A}/\sqrt{-2mH}$ (the classical counterpart of Physlib's quantum commutators), and Bertrand's theorem. For an upstream contribution to Physlib, the general lemma `eq_of_poissonBracket_eq_zero` should move to the Poisson-bracket module, and the work should be split into several smaller pull requests as required by the project's guidelines.

Author contributions. N.L. chose the problem, directed the work and reviewed the results. Claude, an AI system developed by Anthropic, surveyed the existing literature and libraries, designed the statements, wrote the Lean proofs and drafted the manuscript. All statements were checked by the Lean kernel. As an AI system cannot take responsibility for a publication, N.L. is accountable for its content.

References

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[4] Joseph Tooby-Smith. HepLean: Digitalising high energy physics, 2024.